

Eutrobot: An Autonomous Unmanned Surface Vehicle Integrating Real Time Environmental Sensing, AI Assisted Algal Detection, and Phytoremediation for Scalable Freshwater Purification

Parikshitsinh Jadeja*

Navrachana International School

Vadodara, Gujarat, India

jadejaparikshit@outlook.com

Shubhranshu Jha*

Gujarat Public School

Vadodara, Gujarat, India

shubhranshujha110110@gmail.com

*These authors contributed equally to this work and are co-authors in the context of this paper and the conference as declared.

Abstract—Freshwater eutrophication and chemical contamination represent one of the most critical environmental challenges of the 21st century, with approximately 70 to 80% of India's surface water contaminated according to the Central Pollution Control Board (CPCB). Conventional interventions, including Sewage Treatment Plants (STPs) and surface skimmers, are incapable of addressing dissolved pollutants already present in natural water bodies. This paper presents Eutrobot, a fully autonomous, decentralised Unmanned Surface Vehicle (USV) that integrates multi-parameter physicochemical sensing, GPS guided navigation, AI powered algal bloom detection, and adaptive in situ phytoremediation within a single mobile platform. The system continuously monitors 17 water quality parameters, including pH, turbidity, TDS, temperature, and colorimetric heavy metal indicators for lead and mercury stored on an onboard web server. Upon detecting contamination above CPCB thresholds, a dynamic biomass deployment algorithm releases a calculated quantity of *Lemna minor* (common duckweed), utilising a CPCB calibrated weighted average uptake coefficient of 6.9 mg/g/day. Surface algal blooms are detected autonomously via a HuskyLens embedded AI vision module and removed by a servo actuated harvesting mechanism. Field trials at Atladara Lake and Kalali Lake, Vadodara, confirmed stable autonomous navigation and real-world contamination detection, revealing critically elevated lead (20 mg/L) and nitrate (250 mg/L) levels invisible to visual inspection. Results demonstrate that coupling autonomous monitoring with data-driven, adaptive phytoremediation provides a scalable, chemical free approach to actively restoring freshwater bodies beyond the capabilities of existing passive or centralised systems.

Index Terms—eutrophication, phytoremediation, *Lemna minor*, autonomous USV, water-quality monitoring, AI vision, rhizofiltration, embedded systems, freshwater restoration

I. INTRODUCTION

Freshwater ecosystems worldwide are facing an accelerating crisis driven by anthropogenic nutrient enrichment, toxic industrial discharge, and untreated municipal wastewater. The resulting process, eutrophication, generates explosive algal blooms that strip dissolved oxygen from water, causing widespread fish kills, biodiversity collapse, and the release of

cyanotoxins hazardous to human health [1], [2]. The paradox of this crisis is its invisibility: water bodies that appear clear to the human eye frequently harbour biologically lethal concentrations of nitrates, phosphates, and heavy metals.

The socioeconomic consequences are profound. Unsafe water, poor sanitation, and inadequate water management collectively contribute to nearly 1 million deaths annually according to the World Health Organization, with over 475,000 child deaths per year attributable to waterborne disease [2], [3]. In India, the CPCB estimates that 70 to 80% of surface water bodies are contaminated, yet the majority of remediation infrastructure, STPs, floating wetlands, mechanical skimmers, targets only visible or pre discharge pollution. None of these systems is capable of treating chemical contamination already resident in a water body, and none can adapt dynamically to spatially heterogeneous pollution distributions.

Existing aquatic robotic systems largely replicate this paradigm: they monitor passively or remove surface debris, leaving dissolved contaminants and sub surface biological degradation entirely unaddressed. What is required is a system that understands a water body, one that moves intelligently, detects hidden pollution, and actively intervenes at the source.

This paper introduces Eutrobot, an autonomous, decentralised USV designed to close this gap. By unifying real time multi parameter sensing, autonomous GPS/IMU navigation, embedded AI based algal detection, and adaptive in situ phytoremediation, Eutrobot represents one of the world's first integrated freshwater contamination management platforms that acts rather than merely observes.

II. BACKGROUND AND RELATED WORK

A. Phytoremediation with *Lemna minor*

Lemna minor (common duckweed) is among the fastest growing vascular plants on Earth, capable of doubling its biomass within 48 hours under nutrient-rich conditions [4].

Its exceptional surface area to volume ratio enables highly efficient rhizofiltration, the absorption of dissolved contaminants from water through root-zone uptake. Published literature reports nitrogen removal efficiencies of up to 86.9% and phosphorus removal of up to 85.2% from wastewater, with COD reductions exceeding 75% [4]. Nitrate uptake rates of 40 to 310 mg/m²/day and phosphate uptake of 50 to 200 mg/m²/day have been observed in high-contamination environments [5], [6].

Heavy metal remediation by duckweed has also been extensively characterised: lead (Pb) removal efficiencies of 88.8% and cadmium (Cd) of 22.2% are reported, alongside bioaccumulation of polycyclic aromatic hydrocarbons (PAHs), organic dyes, and pharmaceuticals [7], [9]. These properties make *Lemna minor* uniquely suited as a biological intervention agent in a robotic platform.

B. Autonomous Aquatic Systems

Current deployable aquatic robotic platforms fall broadly into three categories: (i) stationary sensor buoys offering high temporal resolution but zero spatial coverage; (ii) remotely operated surface vehicles providing mobility but requiring continuous human oversight; and (iii) autonomous survey USVs focused exclusively on data collection [10], [11]. A critical gap exists in systems that combine monitoring with active remediation, particularly biological, chemical free remediation. Eutrobot is designed specifically to address this gap.

C. AI Assisted Environmental Detection

Machine learning applied to water quality classification and algal bloom detection has shown considerable promise in laboratory settings [12], [13]. Embedded deployment of such models, running inference directly on edge hardware without cloud dependency, remains an active research frontier. The HuskyLens module used in this work represents a practical realisation of embedded AI for real time aquatic monitoring.

III. SYSTEM ARCHITECTURE

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Eutrobot is built around a dual processor architecture: a primary Linux ARM processor (LuckFox Pico) running Ubuntu handles high level navigation, AI signal processing, and autonomous decision making, while a secondary microcontroller (Arduino UNO R4 WiFi) manages real time sensor acquisition and actuator control via I²C communication. The complete hardware and software stacks are summarised in Table I and Table II respectively.

The platform operates in four continuous closed-loop stages:

- 1) **Monitor:** Continuous acquisition of 17 physicochemical parameters.
- 2) **Navigate:** GPS/IMU guided waypoint traversal across the water body.
- 3) **Purify:** Algorithmically calculated *Lemna minor* deployment.

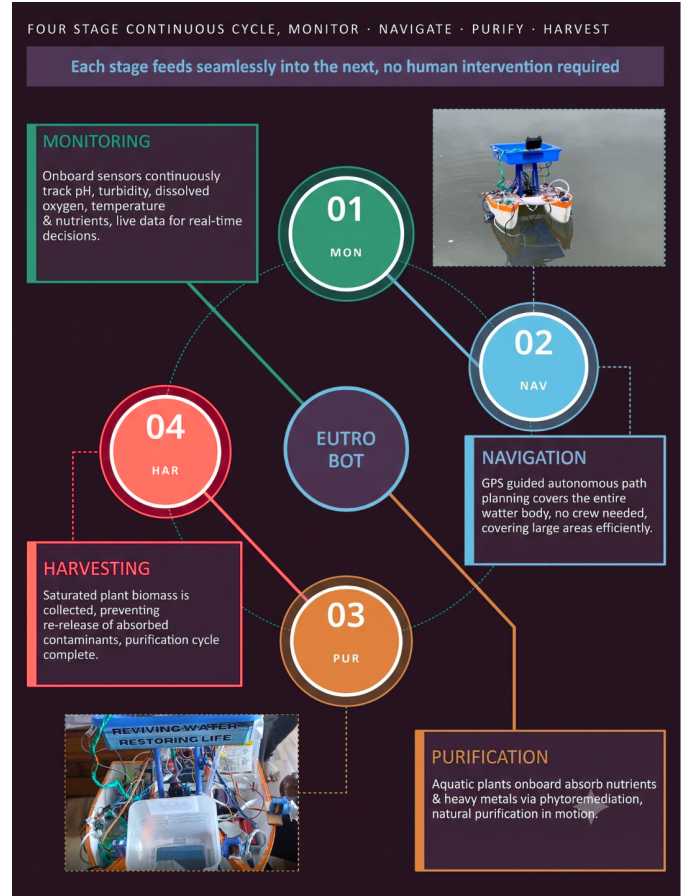


Fig. 1. Eutrobot closed loop autonomous system architecture.

- 4) **Harvest:** AI detected algal bloom skimming via servo mechanism.

A. Hull and Propulsion

The platform is constructed on a 3D printed hydrodynamic hull designed in AutoCAD. Propulsion is achieved via two DC motors coupled with MIT OpenSource toroidal propellers [14], which offer substantially reduced acoustic noise compared to conventional marine propellers, a critical consideration for preserving aquatic fauna behaviour during deployment. Motors are controlled via an L298N dual H bridge driver receiving PWM signals from the Arduino. Motor housings are water sealed as a cost effective alternative to commercial waterproof thrusters.

V. MATERIALS AND METHODS

A. Water Quality Monitoring

Eutrobot monitors 17 physicochemical parameters in real time. Electrochemical sensors measure pH, conductivity (TDS), and turbidity, while a platinum RTD records water temperature. A colorimetric indicator system, calibrated against CPCB drinking water and environmental standards, provides semi quantitative detection of dissolved heavy metals (lead, mercury) and ionic species (nitrates, nitrites, bromine,

TABLE I
HARDWARE COMPONENT SUMMARY

Component	Role
LuckFox Pico (ARM)	Primary compute: navigation, AI, decisions
Arduino UNO R4 WiFi	Sensor I/O, actuator control, I ² C bridge
HuskyLens AI Module	Embedded algal bloom detection
NEO 6M GPS	Real-time position & waypoint navigation
HMC5883L Magnetometer	Heading / orientation (IMU)
pH, Turbidity, TDS Sensors	Acidity, suspended particulate, conductivity
Water Temp Sensor	Thermal stratification detection
Colorimetric System	Lead, mercury, nitrates, nitrites, bromine, hardness
L298N Motor Driver	PWM propulsion control
Relay Module & Pump	<i>Lemna minor</i> dispersal mechanism
ESP32-CAM	Visual documentation & remote monitoring
Toroidal Propellers (MIT)	Silent, disturbance-free aquatic propulsion

TABLE II
SOFTWARE STACK

Component	Function
Ubuntu	OS on LuckFox Pico
Python 3.12	Navigation logic, sensor processing, decisions
Embedded C++	Sensor acquisition, PWM, relay, I ² C
gpsd / pynmea2	NMEA GPS stream parsing
geopy / haversine	GPS bearing & distance calculations
smbus2	I ² C inter processor communication
Matplotlib	Sensor logging & calibration visualisation

hardness, alkalinity). All data are timestamped, geo referenced via GPS, and stored on an onboard web server accessible over Wi-Fi, enabling spatial water-quality mapping across an entire water body during a single autonomous deployment.

B. Autonomous Navigation

The navigation subsystem uses a NEO 6M GPS module in conjunction with an HMC5883L 3 axis digital compass for heading correction. Waypoint coordinates are pre loaded and processed by the haversine bearing algorithm running on the LuckFox Pico. The system computes real time cross track error and issues differential PWM corrections to the two propulsion motors, enabling stable closed loop trajectory following. Navigation logic was validated in a controlled swimming pool environment before open water deployment.

C. AI Assisted Algal Bloom Harvesting

The HuskyLens embedded AI vision module, powered by a K210 RISC V processor with an on chip neural network accelerator, performs real time colour based object recognition to detect surface algal blooms. Detection events trigger a servo actuated skimming mechanism via the Arduino relay system,

physically removing bloom material from the water surface. This mechanism operates entirely on board without cloud inference, ensuring sub-second detection-to-action latency and full operational independence.

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D. Phytoremediation, *Lemna minor* Cultivation and Deployment

E. Phytoremediation, *Lemna minor* Cultivation and Deployment

Lemna minor was selected following systematic evaluation of aquatic macrophytes on the basis of: (i) demonstrated nutrient and heavy-metal uptake efficiency from peer reviewed literature; (ii) rapid exponential biomass growth (doubling time ≤ 48 h); (iii) operational safety for native aquatic ecosystems; and (iv) ease of controlled cultivation and mechanical deployment. Prior to field use, duckweed cultures were maintained under controlled nutrient conditions to validate the reported 48 hour biomass doubling trajectory.

Upon contamination detection, defined as any parameter exceeding CPCB threshold levels, a relay controlled submersible pump disperses duckweed slurry directly into the water column.

1) *Dynamic Biomass Deployment Algorithm*: A key innovation of this work is the replacement of fixed-quantity phytoremediation release with a data-driven dynamic deployment algorithm. The required duckweed mass m_L is calculated as:

$$m_L = \frac{C_p \cdot V_w}{\kappa \cdot t_d} \quad (1)$$

where C_p is the measured pollutant concentration (mg/L), V_w is the estimated treatment volume (L), κ is the CPCB-calibrated weighted average uptake coefficient (6.9 mg/g/day), and t_d is the intended deployment duration (days). This formulation ensures that remediation dose scales continuously with observed contamination, preventing under treatment in high load events and minimising unnecessary biomass introduction in cleaner zones.

F. Field Deployment Protocol

Field trials were conducted across five water bodies: Atladara Lake, Kalali Lake (Vadodara, Gujarat), and river samples from the Narmada, Kshipra, and Ganga. For each deployment, the following protocol was observed:

- 1) Pre deployment sensor calibration against CPCB reference standards
- 2) Autonomous GPS waypoint programming covering the target water body
- 3) Continuous data acquisition at 1 Hz sampling rate
- 4) Real-time threshold comparison and conditional duckweed release
- 5) Post-deployment data extraction and off board analysis via Matplotlib

VI. RESULTS

A. Field Water Quality Assessment

Table III presents measured physicochemical parameters across all tested water bodies. A critical finding is that conventional visual assessment and basic pH measurement systematically failed to predict true contamination levels; water bodies appearing safe harboured dangerous dissolved pollutant concentrations.

TABLE III
MEASURED WATER-QUALITY PARAMETERS ACROSS FIELD SITES

Site	pH	Hardness (mg/L)	Lead (mg/L)	Nitrates (mg/L)
Atladara Lake	8.4	250	20.0	ND
Kalali Lake	7.2	ND	ND	250
Kshipra River	~7.5	Very High	ND	ND
Narmada River	~7.0	Low	Elevated	ND
Ganga River	Stable	Low	ND	ND

*ND = Not detected above threshold; Elevated = above CPCB limit

Atladara Lake recorded a pH of 8.4, alkalinity of 120 mg/L, total hardness of approximately 250 mg/L, and critically, a lead concentration of 20 mg/L, far exceeding the CPCB permissible limit of 0.05 mg/L for surface water. This contamination is attributable to Plaster of Paris idol immersion during Ganesh Chaturthi, which introduces calcium, magnesium, and heavy-metal salts into the water body.

Kalali Lake appeared visually eutrophic with a neutral pH of 7.2, yet harboured nitrate concentrations of 250 mg/L, indicative of severe eutrophic loading capable of sustaining continuous algal bloom cycles and driving dissolved oxygen to near anoxic levels.

Narmada River, appearing pristine to visual inspection, contained elevated heavy-metal concentrations, while the Kshipra exhibited very high hardness despite an acceptable pH. Only the Ganga sampling point showed relatively stable conditions across all parameters. These results collectively confirm that appearance and pH alone are wholly unreliable indicators of water safety.

B. Phytoremediation Efficiency

Laboratory and field validation confirmed that 120 g of *Lemna minor* can purify approximately 0.5 L to 4 L of contaminated water, with purification capacity varying as a function of pollutant concentration and contact time. Duckweed biomass doubled within 48 hours of deployment in nutrient-rich conditions, confirming self-sustaining exponential growth that amplifies remediation capacity over time.

Published removal efficiencies for the key contaminants detected in this study are summarised in the visual analysis framework [4], [6], [7].

C. Autonomous Navigation Performance

Stable autonomous navigation was achieved in both controlled (swimming pool) and open natural water environments.

The integrated GPS/IMU guidance system successfully maintained pathpoint following trajectories under natural surface disturbances, including wind driven currents and boat wash. The system operated continuously without manual intervention throughout all field deployments. Duckweed was released autonomously upon contamination threshold exceedance at both lake sites, confirming end to end closed-loop operation.



Fig. 2. Field validation of autonomous navigation: (top) open-water deployment at Kalali Lake showing eutrophic conditions; (bottom) controlled swimming pool testing confirming propulsion stability and sensor integration.

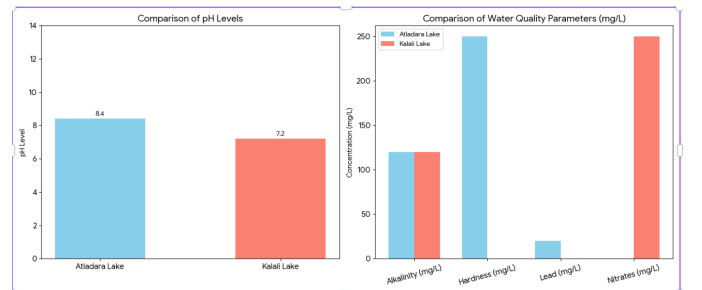


Fig. 3. Comparative water-quality parameters measured by Eutrobot at Atladara and Kalali Lakes, Vadodara. Dashed reference lines indicate CPCB permissible limits.

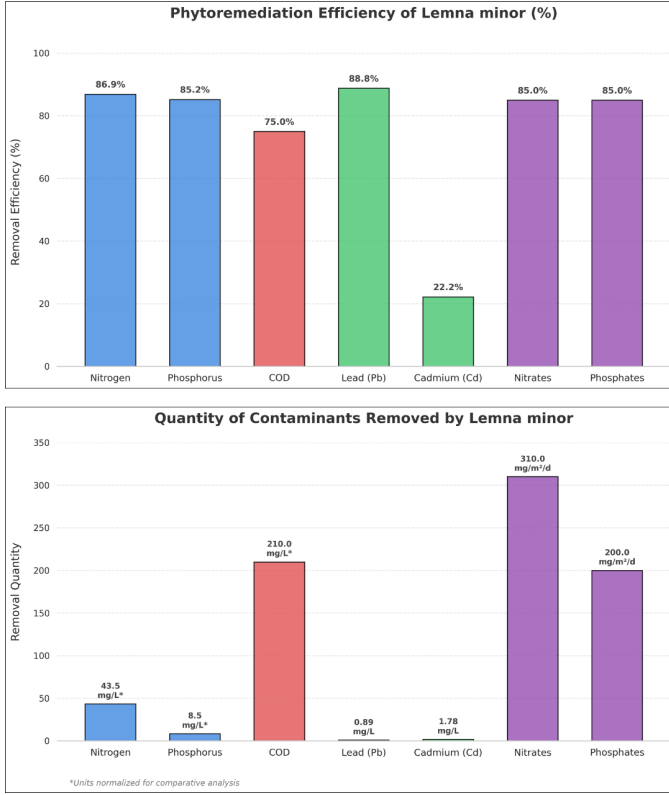


Fig. 4. Phytoremediation efficiency (%) and quantity of contaminants removed by Lemna minor across key pollutant categories.

VII. DISCUSSION

A. Invisible Contamination as a Systemic Failure Mode

The most consequential finding of this study is not a specific contamination measurement, but the systematic divergence between water appearance and water safety. At Atladara Lake, a pH of 8.4, barely outside the neutral range, masked lead concentrations **400× above CPCB permissible limits**. The Narmada, widely regarded as one of India's healthier rivers, contained elevated heavy metals. These findings demonstrate that existing monitoring paradigms based on visual inspection and single-parameter spot checks are **fundamentally insufficient** as public health or ecological tools.

Eutrobot's continuous, multi parameter, spatially resolved monitoring directly addresses this gap. By geo referencing every sensor reading and constructing water quality maps across entire water bodies, the platform generates actionable environmental intelligence that static sensor buoys and manual sampling campaigns cannot provide.

B. Advantages of Adaptive Over Fixed Remediation

Prior phytoremediation deployments in constructed wetlands and experimental systems have universally employed fixed biomass quantities determined in advance [8]. This

approach is inherently inappropriate for natural water bodies where contamination varies spatially by orders of magnitude, as demonstrated by the contrast between pristine-appearing river sections and heavily polluted lake zones in this study.

The dynamic deployment algorithm (Equation 1) represents a fundamental shift: remediation dose is a continuous function of real time sensor data rather than a predetermined constant. This not only improves treatment efficacy in high contamination zones but prevents the introduction of excess biomass that could itself perturb ecosystem balance.

C. System Limitations and Future Directions

Several limitations of the current prototype are acknowledged. The communication range of the onboard Wi-Fi system constrains deployment to water bodies where the platform remains within approximately 100 m of a base station. Future iterations will incorporate **LoRa (Long Range)** wireless communication for multi kilometre autonomous operation across large lakes and river systems.

Quantitative *in situ* validation of phytoremediation efficiency, comparing pre and post-treatment samples from the same GPS coordinates, was not completed in this study phase due to temporal constraints. Long term controlled deployment trials are planned. Solar energy integration will extend operational endurance beyond the current Li-Ion battery limitation, enabling multi day continuous deployments.

Additional planned enhancements include: (i) embedded edge AI for real-time eutrophication severity classification; (ii) floating aerator modules to directly raise dissolved oxygen levels; and (iii) multi-unit swarm deployment for parallel monitoring of large water bodies.

Key Findings at a Glance

- 17 parameters monitored in real time across 5 water bodies.
- Lead at 20 mg/L detected at Atladara Lake, 400× above CPCB limit, in visually normal water.
- Nitrates at 250 mg/L at Kalali Lake driving visible eutrophication.
- *Lemna minor* biomass doubled within 48 h in field conditions.
- Stable autonomous GPS navigation achieved in open water.
- Dynamic biomass algorithm replaces fixed dose phytoremediation.
- Fully autonomous closed-loop operation with zero human intervention.

VIII. CONCLUSION

This study has demonstrated that freshwater pollution is substantially more severe and hidden than surface level observations indicate. Eutrobot, an autonomous, decentralised USV integrating real time multi parameter sensing, GPS guided navigation, AI-assisted algal harvesting, and adaptive *in situ* phytoremediation, provides a scalable, chemical free response to this challenge that existing infrastructure fundamentally cannot deliver.

Field validation across five water bodies confirmed the platform's ability to detect hidden contamination, navigate

autonomously in natural conditions, and deploy biological remediation agents in response to real time data. The dynamic biomass deployment algorithm moves water treatment from a passive, infrastructure dependent paradigm to an active, data driven ecological intervention model.

Eutrobot demonstrates how the convergence of robotics, embedded AI, and nature-based solutions can produce environmental technologies that do not merely monitor the crisis but actively address its root causes, bringing life back to water bodies rather than simply delaying their loss.

Significance Statement: By integrating environmental sensing, AI assisted pollution identification, phytoremediation, and active purification within a single mobile robotic system, this study demonstrates one of the world's first approaches towards freshwater contamination management that extends beyond the capabilities of traditional monitoring systems and surface cleaning technologies, by targeting hidden contamination within water bodies.

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REFERENCES

- [1] Central Pollution Control Board (CPCB), *Annual Report on Water Quality of Rivers and Lakes in India*, Ministry of Environment, Forest and Climate Change, New Delhi, 2022.
- [2] World Health Organization, *Drinking-Water Fact Sheet*, WHO, Geneva, 2023. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/drinking-water>
- [3] UNICEF, *Water, Sanitation and Hygiene: Progress and Challenges*, UNICEF, New York, 2022.
- [4] S. A. Teixeira *et al.*, "Growth and Nutrient Removal Efficiency of Duckweed (*Lemna minor*) in Wastewater and Landfill Leachate," *PLOS ONE*, vol. 14, no. 9, e0221755, 2019.
- [5] "Nutrient Removal by *Lemna minor* from Wastewater," *Int. J. Sci. Res. & Academic*, vol. 13, no. 2, 2024.
- [6] M. Davoodikia, K. R. Tavabe, and A. M. Moradi, "Evaluation of the common duckweed (*Lemna minor*) phytoremediation on nitrate, phosphate, cadmium and lead parameters reduction from aquaculture wastewater," *Aquaculture Sciences*, vol. 9, no. 1, pp. 192–202, 2021.
- [7] E. A. Mahdi Al-Nabhan, "Removal Efficiency, Accumulation and Biochemical Response of *Lemna minor* Exposed to Some Heavy Metals," *IOP Conf. Ser.: Earth Environ. Sci.*, vol. 1060, 012037, 2022.
- [8] A. F. Hasaballah, T. A. Hegazy, and D. A. Elemam, "Performance evaluation of *Pistia stratiotes* and *Lemna minor* for wastewater phytoremediation: optimum conditions for pilot scale," *Int. J. Environ. Sci. Technol.*, vol. 21, pp. 467–480, 2023.
- [9] "The Potential of Common Duckweed (*Lemna minor*) in Phytoremediation of Phenanthrene and Pyrene," *Environ. Eng. Res.*, 2023.
- [10] "Advanced Sensor Technologies for Water Quality Monitoring," *Sensors*, vol. 21, no. 22, 7637, 2021.
- [11] "Sensor-Based Intelligent Monitoring Systems for Water Quality," *Sensors*, vol. 21, no. 4, 1102, 2021.
- [12] "AI, Machine Learning and Smart Technologies for Water Treatment," *Water*, vol. 17, no. 11, 1641, 2022.
- [13] D. Pal *et al.*, "Detecting Chemical Contaminants in Water Using AI," in *Smart Water Technologies*, Taylor & Francis, 2024.
- [14] MIT OpenSource Toroidal Propeller Design. [Online]. Available: <https://web.mit.edu>
- [15] "Advanced Environmental Monitoring Approaches for Water Ecosystems," *Front. Environ. Sci.*, vol. 12, 1336088, 2024.
- [16] "Role of Aquatic Plants in Phytoremediation," *Plants*, vol. 12, no. 3, 589, 2023.